LEVEL 6

MATHEMATICS FOR COMPUTER SCIENCE

JKUAT TVET INSITTUTE – KAREN CAMPUS
WRITTEN ASSESSMENT
CAT 2

NB ANSWER ALL QUESTIONS

1. Solve $5x - 6 = 3x - 8$ (4 Marks)

2. Solve this system of equations graphically. (6 Marks)

$$4x - 6y = 12$$

$$2x + 2y = 6$$

3. Describe the following terms as applied in probability and statistics. (10 Marks)

   i. Population

   ii. Data

   iii. Variables

   iv. Sample space

   v. Event

4. Given the following data calculate geometric mean: (5 Marks)

   130, 135. 140, 145, 146, 148, 149, 150, 157

5. The marks (out of 50) scored by 30 students in a statistics test are given below:

   32, 18, 25, 29, 41, 20, 23, 35, 38, 22,

   26, 33, 17, 45, 28, 30, 19, 24, 21, 27,

   31, 36, 40, 34, 37, 39, 44, 43, 42, 46

   a) Construct a **grouped frequency distribution table** using a **class width of 5**, starting from the lowest mark. (5Marks)

b) Use the grouped data to compute the **mean**.(5Marks)

c) Use the grouped data to compute the **variance**.(5Marks)

6. A company produces two types of animal feeds: **Type A** and **Type B**.

Each **bag of Type A** requires:

- 3 kg of maize,
- 2 kg of soya,
- 1 kg of fishmeal.

Each **bag of Type B** requires:

- 4 kg of maize,
- 1 kg of soya,
- 2 kg of fishmeal.

In one day, the company used:

- 180 kg of maize,
- 70 kg of soya, and
- 60 kg of fishmeal.

a) Let the number of bags of Type A be x and Type B be y
Form a system of linear equations representing the situation above.(2Marks)

b) Represent the system in **matrix form**(3Marks)
AX=B

c) Use **matrix methods (inverse of a matrix)** to find the number of bags of each type produced that day.(5Marks)

21

JKUAT CONTINOUS ASSESSEMENT TESTS

NAME: KIAMIRA JOSHUA NYAGA TVET

REG NO: TDC113-C002-0007/2024

COURSE: Diploma In Computer Science YEAR OF STUDY: 1

SHEET NO:

UNIT CODE: TITLE: Mathematics For Computer Science

DATE:

NOTE: This stationery will be used for Continuous Assessment work only

It will be a breach of examination regulations to use it otherwise.

1. $5x - 6 = 3x - 8$

$+3x + 6 \quad +6 - 3x$

$5x + 3x = -8 + 6$

$8x = -2$

$x = -0.25$

2. $4x - 6y = 12$

x	0	3
y	-2	0

$2x + 2y = 6$

x	0	3
y	3	0

2

3) i) Population - Refers to number of items, objects, people within an area

ii) Data - Refers to records that are collected with an area of a population

iii) Variables - Refers to values that keeps on changing with time or due to various reasons that is they are not constant

iv) Sample

v) Event - Refers to an action that occurs in a particular environment at a time.

i. 130, 135, 140, 145, 146, 148, 149, 150, 157

$n = 9$

$G.M = \sqrt[n]{x_1 \times x_2 \times x_3 \times x_n}$

$G.M = \sqrt[9]{130 \times 135 \times 140 \times 145 \times 146 \times 148 \times 149 \times 150 \times 157}$

$G.M = \sqrt[9]{2.701 \times 10^{19}} = 144.23$

5

5) ~~Class f~~
~~20 - 24~~
~~25 - 29~~
~~30 - 34~~
~~35 - 39~~
~~40 - 44~~
~~45 - 49~~

Class f
15 - 19 3
20 - 24
25 - 29
30 - 34
35 - 39
40 - 44
45 - 49

5) a)

Class	f	Tally	x	$t = x - A$ (27)	ft	ft^2
15-19	3	III	17	-10	-30	300
20-24	5	~~IIII~~	22	-5	-25	125
25-29	5	~~IIII~~	27	0	0	0
30-34	5	~~IIII~~	32	5	25	125
35-39	5	~~IIII~~	37	10	50	500
40-44	5	~~IIII~~	42	15	75	1125
45-49	2	II	47	20	40	800
	$\Sigma f = 30$				$\Sigma ft = 125$	$\Sigma ft^2 = 2975$

5

b) Mean $A = 27$

$\bar{x} = \frac{\Sigma ft}{\Sigma f} + A$ Assumed mean

$\bar{x} = \frac{135}{30} + 27$

$\bar{x} = 4.5 + 27$

$\bar{x} = 31.5$

5

c) Variance

$\text{Variance} = \frac{\Sigma ft^2}{\Sigma f} - \left(\frac{\Sigma ft}{\Sigma f}\right)^2$

$\text{Variance} = \frac{2975}{30} - \left(\frac{135}{30}\right)^2$

$\text{Variance} = 78.9166$

$= 78.92$

15

TDL 112 - 1002 - 0003 12 01H

a) $A = x$ $\quad$ $B = y$

$$\begin{array}{c|ccc} & M & S & F \\ \hline x & 3 & 2 & 1 \\ y & 4 & 1 & 2 \end{array} \times \begin{pmatrix} 180 \\ 70 \\ 60 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$3x + 2x + x = 18$

$A = \{ 3 \times 180 + 2 \times 70 \times$

6 a) $3x + 4y = 180$ $\quad$ $x + y = 310$

$2x + y = 70$ $\quad$ $5x +$

$x + 2y = 60$

$y = 70 - 2x$

$3x + 4y = 180$

$3x + 4(70 - 2x) = 180$

$3x + 280 - 8x = 180$

$-5x =$

6a) $A = x$ $\quad$ $B = y$

$x + y = 310$

$6x + 7y = 310$

$180 + 70 + 60 = 310$

$3 + 2 + 1 = 6$

$4 + 1 + 2 = 7$

$x + y = 310$

$6x + 7y = 310$

b) $A(X) = B$

$$\begin{pmatrix} A \\ B \end{pmatrix} \begin{pmatrix} M & S & F \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 180 \\ 70 \\ 60 \end{pmatrix}$$